

Bryce Verberne

bryceverberne.com • Active DoD TS Clearance • [linkedin.com/in/bryce-verberne](https://www.linkedin.com/in/bryce-verberne) • github.com/BryceVerberne

EDUCATION

B.S, Computer Science
Arizona State University, Tempe, AZ

Dec 2025
3.74 GPA

SKILLS

Programming Languages: C++, Python, C, Java

Tools & Technologies: RHEL, Kubernetes, Docker, Git, GTest, Jira, GDB, MongoDB, SQLite, RTOS, Microcontrollers, OpenC3

Technical Areas: Distributed Systems, Multithreading, Containerization, Unit and Integration Testing, Embedded Systems, SysML

EXPERIENCE

Northrop Grumman

Software Engineer I

Jan 2026 – Present

- Develop scalable C++ services in RHEL for a distributed, containerized missile defense application on an 11-engineer team.
- Implement time-sensitive service communication and validate C++ services with GTest, static analysis, and Kubernetes testing.
- Selected as a Data Transfer Agent (DTA) and secure-area primary, coordinating approved transfers, access, and security.
- Earned a Northrop Grumman BRAVO Award for rapid, dependable delivery, and program-wide contributions.

Software Engineering Intern

Jun 2025 – Jan 2026

- Supported NG's Missile Defense division by developing a multithreaded C++ pipeline streaming Cameo simulations into AFSIM.
- Developed MongoDB interface to track changes, map and validate jobs, and queue jobs for execution, storing results in SQLite.
- Performed unit, integration, stress, and regression tests to ensure system reliability and performance.
- Built HTML/CSS/JS GUI for real-time job tracking and authored SysML sequence/class diagrams in Cameo.
- Followed Agile workflows with Jira, using Git and Bitbucket under the OUTMATCH IRAD program.

Sun Devil Satellite Laboratory Club

ASCEND: Deputy Software Lead

Aug 2024 – May 2025

- Co-led a 10-member software team using Scrum, Jira, and GitHub to design and develop flight software (C++), ground software (Python), and data processing tools for a NASA-funded high-altitude balloon project.
- Established weekly sprints and retrospectives, identified bottlenecks, and mentored team members to drive continuous advances.
- Refactored the legacy codebase to an OOP design in C++, streamlining new hardware integration for robust flight operations.
- Reduced data footprint by 67% by migrating from CSV-based logging to a packet-based system, improving storage efficiency.
- Enhanced SPI flash memory by designing onboard file system and payload commands for data management.
- Built a packet decoder for a multi-threaded Python ground software pipeline, enabling high-throughput real-time data processing.

Coconut CubeSat: Software Engineer

Apr 2024 – Jan 2025

- Worked on a 7-member team to design and develop the software system for the Coconut CubeSat, a 1U satellite project.
- Wrote Python scripts to parse and route telemetry packages, facilitating satellite-to-ground communication.
- Integrated OpenC3 into ground software for hardline and radio communication using CCSDS protocols.

NASA L'SPACE MCA

Deputy Project Manager

Mar 2024 – Apr 2024

- Co-led a team of 12 to design a conceptual Mars rover, overseeing milestone reviews including MCR, SRR, MDR, and PDR.
- Spearheaded programmatic sub-team, tracking budget, schedule, and risks.
- Facilitated monthly iterative phases with specific goals and deliverables, conducting regular reviews and incorporating feedback.
- Bridged Science and Engineering teams for effective cross-functional collaboration and risk management.

Computer Hardware Engineer

Jan 2024 – Apr 2024

- Developed the Command and Data Handling (CDH) Subsystem of a conceptual Mars rover design, entailing the telecommunications, data computing, and software architecture subassemblies.
- Developed lower-level requirements and conducted trade studies to select sub-components.
- Identified risks, implemented redundancy, and determined mass, volume, and power specifications.
- Created software architecture flowchart and developed manufacturing and procurement plans.